

Suparnojit Sarkar

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EDUCATION

Heritage Institute of Technology

Bachelor of Technology in Computer Science

Kolkata, India

2022-2026

EXPERIENCE

Glance AI(InMobi group)

Applied Scientist

February 2026 – Present

Bengaluru, India

- Worked on conversion of Z Image base model from T2I to I2I using training free techniques
- Worked on realism for image generation using LoRAs applied on Flux Klein and Qwen Image
- Working on Multi Tryoff-Flatlay-VTON PoC along with moderation handling at inference time

Cohere Labs

Research Collaborator

February 2026 – Present

Remote

- Working on Tiny Aya Vision and Safety Representations across Languages on Tiny Aya models as a part of Expedition Aya project
- Implemented a cross-modal weight merging strategy by interpolating multimodal and base LLM weights with a tunable alpha, enabling recovery of text-only capabilities while preserving visual grounding performance
- Calculated refusal rate across all languages on Tiny Aya variants using LLM as a Judge (Command A) on X-Safety benchmark

CINT Lab, Indian Institute of Science

Project(Gen AI/ML) Intern

July 2025 – August 2025

Bengaluru, India

- Implemented end to end pipelines for image/video generation using Flux 1Dev/Qwen Image and Wan 2.2(text to image and image to video)
- Fine tuned Flux.1-Dev (LoRA) for Studio Ghibli image generation with 80% accuracy
- Zero-Shot, Few-Shot and CoT Prompt Engineering for proper image/video generation on ComfyUI

VIML Lab, Indian Institute of Technology Mandi (IIT Mandi)

ML/CV Intern

December 2024 – January 2025

Mandi, India

- Worked on Semantic Segmentation of Unmanned Aerial Vehicle Remote Sensing Images under the guidance of Prof.Dinesh Singh
- Training and inferring of UAVid dataset using 2 different models (SegFormer,UnetFormer) models for image segmentation using Pytorch-Lightning and achieved F1 score of 0.65

PROJECTS

Fashion-Search | *Python, PyTorch, Hugging Face Transformers, ChromaDB, VLM*

- Built hybrid fashion retrieval using CLIP embeddings, K-means color extraction, and zero-shot scene classification with weighted re-ranking, enhanced with hierarchical indexing and inverted indices for 80% candidate filtering.
- Designed semantic caching layer achieving 75-85% cache hit rates, enabling sub-20ms response times for repeated and similar queries.
- Fine-tuned CLIP ViT-L/14 with contrastive learning, hard negative mining, and gradient checkpointing to optimize for low resource while improving fashion-specific query precision by 15-20%.

LegalAI | *Python, Agno AI, Streamlit, ChromaDB*

- Developed a multi-agent legal document analysis system with 4 specialized AI agents (LegalAdvisor, ContractAnalyst, LegalStrategist, Team Lead) that process PDF contracts to deliver structured risk assessments, compliance checks, and concise actionable insights via a Streamlit-based interactive UI.
- Integrated ChromaDB with Gemini embeddings for semantic search across legal documents, featuring configurable chunking (1–5000 tokens, 200-token overlap), dual-party contract analysis, risk-based clause highlighting, and real-time chat functionality.

Polygon Colorizer | *PyTorch, Diffusers, and Weights and Biases*

- Built a diffusion-based conditioned U-Net (UNet2DConditionModel) to colorize grayscale polygon images using CLIP text embeddings from color descriptions.

- Leveraged PyTorch, Diffusers, and Weights and Biases for training, experiment tracking, and visualization.

SLM for English Stories | *PyTorch, Transformers, SLM Pre Training*

- Built a 30M-parameter Small Language Model (SLM) for short English story generation, designing the pre training pipeline from data preprocessing to autoregressive text generation.
- Implemented and evaluated transformer-based text generation workflow, demonstrating hands-on expertise in lightweight LLM/SLM training, inference, and qualitative evaluation.

ACHIEVEMENTS

Secured **first place** out of **3000** participants at one of Eastern India's largest hackathons, HackNITR 5.0 coordinated by DSC NIT Rourkela

Mentored and judged AI/ML projects at one of Eastern India's largest hackathons, HackNITR 7.0 coordinated by DSC NIT Rourkela

Top 30 at Hack Heritage 2023 hackathon arranged by Heritage Institute of Technology, Kolkata

Top 2.5% (ranked 2013) among 82,796 participants at Amazon ML Challenge

PUBLICATIONS

IndiaCultureVL-Bench: Cultural Recognition and Prompt-Induced Collapse in Open Vision-Language Models(**in review at an EMNLP 2026 workshop as solo first author**)

IndicMedDialog: A Parallel Multi-Turn Medical Dialogue Dataset for Accessible Healthcare in Indic Languages (**accepted at BioNLP workshop at ACL 2026 as shared first author**)

MedAidDialog: A Multilingual Multi-Turn Medical Dialogue Dataset for Accessible Healthcare (**accepted at MeLLM workshop at ACL 2026 as shared first author**)

Improving Biomedical Relation Classification with Error-Aware Remediation and Curriculum Learning via a Teacher-Student Framework (**in review at ARR May 2026 as second author**)

TECHNICAL SKILLS

Languages/Frameworks: Python, SQL, PyTorch, FastAPI, Agno AI, Crew AI

Tech/Dev Tools: Git, Docker, Dune, HuggingFace, ComfyUI, AI Toolkit, Unsloth, Streamlit, ChromaDB, ArgoCD

Libraries/Others: Fine Tuning, Prompt Engineering, Quantisation, Segmentation, RAG, LLM, Agentic AI